



Original Article

Prognostic impact of heart rate during atrial fibrillation on clinical outcomes in elderly non-valvular atrial fibrillation patients: ANAFIE Registry sub-cohort study[☆]

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ABSTRACT

Background: Elderly patients with atrial fibrillation (AF) are at a higher risk for all-cause mortality and heart failure. Rate control is an essential component in AF management. This exploratory study assessed the relationship between resting heart rate during AF at baseline and clinical outcomes in Japanese elderly non-valvular AF (NVA) patients, using the All Nippon AF In the Elderly Registry (ANAFIE) dataset.

Methods: This sub-cohort included patients who agreed to participate and presented with AF at enrollment in the ANAFIE study. They were categorized into six groups according to the resting heart rate during AF. Outcomes included 2-year cumulative incidences of stroke/systemic embolic events (SEE), ischemic stroke, major bleeding, cardiovascular (CV) events, CV death, all-cause death, and net clinical outcome, a composite of stroke/SEE, major bleeding, and all-cause death.

Results: Of the 8292 patients included in this sub-cohort (paroxysmal, 1496; non-paroxysmal, 6796), 90 % of patients were using anticoagulants. Higher heart rate was more frequently reported in women and in patients with paroxysmal AF and was associated with increased use of direct oral anticoagulants (DOACs) and antiarrhythmic drugs. Heart rate ≥ 110 beats per minute (bpm) was associated with a significantly higher incidence of cardiac events and numerically higher incidences of CV death and all-cause death compared with a heart rate of 60 to <80 bpm, all of which were driven by an increased risk in patients with non-paroxysmal AF. Hazard ratios by the type of anticoagulant for each clinical outcome were comparable across all heart rate categories, indicating no significant interactions.

[☆] Trial registration: UMIN000024006.

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Conclusions: Elderly Japanese patients with non-paroxysmal NVAf and a heart rate ≥ 110 bpm have an increased risk of cardiac events. There was no interaction between heart rate category and the relative risk of adverse clinical events in patients taking DOACs compared with those taking warfarin.

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Introduction

Elderly patients with atrial fibrillation (AF) are known to be at a higher risk for all-cause mortality and heart failure (HF) [1]. The main therapeutic target in patients with AF is maintenance of sinus rhythm; however, the Atrial Fibrillation Follow-up Investigation of Rhythm Management (AFFIRM) trial, a randomized, prospective study that compared the long-term effects of rate control with those of rhythm control [2], and the Japanese Rhythm Management Trial for Atrial Fibrillation (J-RHYTHM) [3] showed no difference in all-cause death, cardiovascular (CV) death, or hospitalization for HF exacerbation between patients who underwent heart rate adjustment therapy or sinus rhythm maintenance therapy. This would also be expected to hold true in elderly AF patients, in whom pharmacological/non-pharmacological sinus rhythm maintenance is difficult.

Treatment guidelines recommend rate control as an integral part of AF management; lenient rate control is an acceptable initial approach unless symptoms call for stricter rate control [4,5]. The Outcomes Registry for Better Informed Treatment of AF (ORBIT-AF) noted higher all-cause mortality in AF patients with both lower and higher heart rates [6]. In the Rate Control Efficacy in Permanent Atrial Fibrillation (RACE) II randomized controlled trial of permanent AF patients, there were no differences in composite clinical events, New York Heart Association class, or hospitalizations between the strict (target heart rate <80 bpm at rest and <110 bpm during moderate exercise) and lenient (heart rate target <110 bpm) arms [7]. This was further substantiated by an analysis from the AFFIRM and RACE trials [8,9].

The heart rate in AF is frequently faster than that in sinus rhythm. This elevated heart rate can adversely affect cardiac function, leading to a tachycardia-induced cardiomyopathy, which may be reversed by controlling the heart rate either with pharmacological intervention or atrioventricular node ablation using radiofrequency current [10]. Rate control, often the preferred therapy for AF, is not inferior to rhythm control for the prevention of death and morbidity from CV causes [11].

Although studies report a relationship between heart rate in AF and all-cause death or CV death [6,9,12], there is a dearth of data determining the relationship between heart rate and stroke/systemic embolic event (SEE) particularly in elderly patients with AF. Hence, the objective of this exploratory sub-cohort study was to examine the relationships between resting heart rate in AF at baseline and clinical outcomes in Japanese elderly non-valvular AF (NVAf) patients, using the dataset of the All Nippon AF In the Elderly (ANAFIE) Registry.

Methods

Study design

The ANAFIE Registry was a multicenter, prospective, observational cohort study with a 2-year follow-up of Japanese patients aged ≥ 75 years with a definitive diagnosis of NVAf. The details of the ANAFIE study design have been previously described in full [13].

The ANAFIE Registry (registered with the identifier UMIN000024006) was approved by the ethics committees of the participating centers. The study was conducted in accordance with the Declaration of Helsinki and complied with local regulatory requirements for registries and ethical guidelines in Japan. Written informed consent was provided by all participants (or family members for patients with communication disorders,

such as aphasia or cognitive impairment); patients were able to withdraw from participation at any time.

Patients

Detailed eligibility and exclusion criteria for the ANAFIE Registry have been previously published [13]. Enrolled patients were elderly ambulatory patients (≥ 75 years) diagnosed with NVAf by electrocardiogram who could visit the study sites. Patients were excluded if they were already participating or planning to participate in an interventional study; or who had a definite diagnosis of mitral stenosis; artificial heart valve replacement (either mechanical or tissue valve prostheses); very recent history of CV events including stroke, myocardial infarction, cardiac intervention for ischemic disease other than myocardial infarction, HF requiring hospitalization, or any bleeding leading to hospitalization within 1 month prior to enrollment; life expectancy of <1 year; or when participation was deemed inappropriate by treating physicians.

The cohort described here included patients who had any type of AF at enrollment in the ANAFIE Registry and who consented to participate in this sub-cohort study to examine the relationship between heart rate and clinical outcomes.

Study measures and endpoints

For this subgroup analysis, resting heart rate in AF, including paroxysmal and non-paroxysmal AF, was measured in all registered patients using a 12-lead electrocardiogram at the time of enrollment and informed consent provision (i.e. baseline), with a tolerance of ± 60 days. The patients were categorized into six groups according to the observed heart rate: <50 bpm; 50 to <60 bpm; 60 to <80 bpm; 80 to <100 bpm; 100 to <110 bpm; and ≥ 110 bpm.

Data on oral anticoagulant (OAC) exposure [including no exposure (No-OAC), direct OAC (DOACs) exposure, and warfarin exposure] were also collected. The study outcomes were the 2-year cumulative incidences of stroke/SEE, ischemic stroke, major bleeding, CV events, CV death, all-cause death, and net clinical outcome (a composite of stroke/SEE, major bleeding, and all-cause death). Outcomes were evaluated in subgroups stratified by heart rate and in subgroups stratified by OAC use. The analysis of outcomes by OAC use was limited to patients with a heart rate of 60 to <80 bpm. This rate was selected as it is considered the optimal heart rate in healthy individuals and accounted for almost half (49.2 %) of subjects included in this study.

Statistical analysis

The planned sample size for this sub-cohort study was set at 6000 patients. For this analysis, frequency tables were created for categorical variables, and summary statistics were calculated for continuous variables. Where applicable, the frequency or incidence of events and their 95 % confidence intervals (CIs) were calculated. For categorical variables, p -values were calculated using the chi-squared test. For continuous variables, p -values were calculated using analysis of variance. No imputations were made for missing data, which were not included in the analyses.

The cumulative incidence rates and 95 % CIs of clinical outcomes at 2 years of follow-up were evaluated by Kaplan–Meier analysis

for subgroups stratified by baseline heart rate in AF. In a multivariate analysis, hazard ratios (HRs) and corresponding 95 % CIs were calculated using Cox proportional hazards models, adjusted for known confounding factors, for comparisons of all heart rate subgroups to the reference subgroup (60 to <80 bpm).

The tests were two-sided, and a p -value of <0.05 was considered statistically significant. All statistical analyses were conducted using SAS version 9.4 or higher (SAS Institute, Tokyo, Japan).

Results

Patients

Of the 32,275 patients enrolled in the ANAFIE Registry, 15,591 were included in this sub-cohort study. This included 6766 patients with paroxysmal AF and 8825 with non-paroxysmal AF. The final analysis set included 8292 patients who had electrocardiogram tracings recorded during AF at baseline (paroxysmal, $n = 1496$; non-paroxysmal, $n = 6796$). In the final analysis set, 58.3 % of patients used antiarrhythmic agents, 50.9 % were on rate control therapy, and 10.8 % were on rhythm control therapy [13]. Fig. 1 illustrates the distribution of baseline heart rates in AF recorded at the time of enrollment. Mean $\pm$ standard deviation heart rate was 76.3 ± 17.3 bpm in the final analysis set and was similar between patients with paroxysmal AF (79.8 ± 22.0 bpm) and non-paroxysmal AF (75.5 ± 15.9 bpm). The baseline characteristics of the patients are shown in Table 1. Compared with the lower heart rate subgroups, patients with a higher heart rate were more likely to be female, have lower CHADS₂ and HAS-BLED scores, were more likely to have undergone catheter ablation and were more likely to have a lower prevalence of comorbidities, including previous myocardial infarction, HF, and cerebrovascular disease (all $p < 0.05$). In addition, the proportion of patients with paroxysmal AF increased remarkably with increasing heart rate, from 15.7 %–18.6 % in the heart rate < 100 bpm category to 25.6 % in the 100 to < 110 bpm category, and to 42.1 % in the ≥ 110 bpm category. At baseline, patients were receiving various non-pharmacologic therapies for AF, but the most common in all heart rate categories were catheter ablation and pacemaker. The majority of patients in all heart rate subgroups were using anticoagulants, although DOACs were used more commonly in higher heart rate groups ($p < 0.001$), whereas warfarin was used more commonly in lower heart rate groups ($p < 0.001$). Both rhythm control and rate control drugs were used more commonly in higher heart rate groups ($p < 0.001$).

Online Table 1 summarizes the baseline characteristics of patients with paroxysmal and non-paroxysmal AF. Patients with non-paroxysmal AF

were more commonly male, had lower systolic blood pressure, and higher clinical risk scores (CHADS₂ and HAS-BLED) than those with paroxysmal AF, and were more likely to have renal dysfunction and heart failure. Conversely, the use of rhythm control drugs was low, the use of rate control drugs was high, the use of antiplatelet drugs was low, and the use of anticoagulants was high in patients with non-paroxysmal AF vs paroxysmal AF.

Outcomes

Kaplan–Meier curves for each clinical outcome in subgroups defined by heart rate during AF are shown in Fig. 2. The incidence of CV events, CV death, all-cause death, and net clinical outcomes tended to increase with increasing heart rate (Table 2). Fig. 3 depicts the adjusted HR (aHR) for each clinical outcome, namely, stroke/SEE, ischemic stroke, major bleeding, cardiac events, CV death, and all-cause death, for the six heart rate groups, with a heart rate of 60 to < 80 bpm as the reference. In a multivariate analysis, patients with a baseline heart rate ≥ 110 bpm during AF had a significantly higher incidence of cardiac events (aHR, 1.49; 95 % CI, 1.07–2.08; $p = 0.019$) and numerically higher incidences of CV death and all-cause death compared with the reference subgroup (60 to < 80 bpm). This was driven entirely by an increased risk in the patients with non-paroxysmal AF (cardiac events: aHR, 1.82; 95 % CI, 1.23–2.67; CV death: aHR, 2.11; 95 % CI, 1.07–4.18; all-cause death: aHR, 1.53; 95 % CI, 1.00–2.34), as no increased risk was observed in patients with paroxysmal AF (Fig. 4). As an exploratory result, the incidence of ischemic stroke was significantly higher, and the incidences of cardiac events and CV events were significantly lower in patients with paroxysmal AF and a heart rate of 80 to < 100 bpm compared with the reference category (Fig. 4).

In patients with a baseline resting heart rate of 60 to < 80 bpm, a significantly higher incidence of stroke/SEE (HR, 2.08; 95 % CI, 1.06–4.07; $p = 0.033$); ischemic stroke (HR, 2.57; 95 % CI, 1.23–5.36; $p = 0.012$); and net clinical outcomes (HR, 1.62; 95 % CI, 1.13–2.33; $p = 0.009$) was observed in patients who were not receiving OACs compared with patients treated with warfarin (Table 3). The incidence of these events was further reduced in patients taking DOACs compared with those taking warfarin, although this did not reach statistical significance (Table 3). Across all heart rate groups, there was no interaction between the heart rate category and HRs between the types of anticoagulants used. The HR for No-OAC vs warfarin was not calculated because of the small number of cases in other heart rate categories. The results for the entire subcohort were similar to those of patients in the 60 to < 80 bpm category, although there was no significant difference; the No-OAC group tended to have a higher risk of stroke/SEE and ischemic stroke. HRs for DOAC vs warfarin consistently tended to be < 1 across other heart rate categories and subcohorts for which HRs could be calculated, and no interaction was observed.

Univariate and multivariate analyses in the Cox proportional hazards model according to the type of anticoagulant used (excluding off-label doses of DOACs) were conducted. The results are shown in Online Table 2.

Discussion

In this sub-cohort study of the ANAFIE Registry population, the main finding was that patients with non-paroxysmal AF and a baseline resting heart rate ≥ 110 bpm during AF had a significantly increased risk of cardiac events (myocardial infarction, cardiac intervention for ischemic disease other than myocardial infarction, HF requiring hospitalization, or CV death), CV death or all-cause death compared with patients with a heart rate of 60 to < 80 bpm. However, heart rate was not an independent risk factor for these events in patients with paroxysmal AF.

Among patients with a heart rate of 60 to < 80 bpm, those not taking OACs had a significantly higher incidence of stroke/SEE, ischemic stroke,

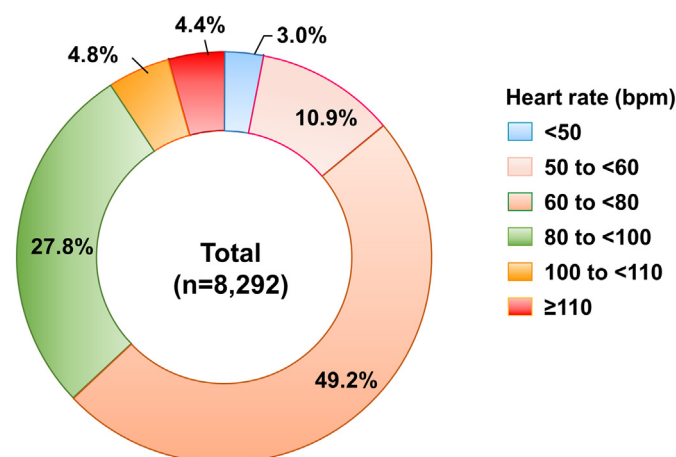


Fig. 1. Distribution of patients according to resting heart rate in atrial fibrillation. bpm, beats per minute.

Table 1
Background characteristics of patients by resting heart rate in AF at the time of enrollment.

Baseline characteristics	Heart rate (bpm); N = 8292						p-Value
	<50 (n = 246)	50 to <60 (n = 900)	60 to <80 (n = 4079)	80 to <100 (n = 2305)	100 to <110 (n = 399)	≥110 (n = 363)	
Sex, male	190 (77.2)	633 (70.3)	2461 (60.3)	1219 (52.9)	188 (47.1)	152 (41.9)	<0.001
Age, years	81.1 ± 4.8	81.7 ± 4.9	81.7 ± 4.9	81.8 ± 4.8	81.9 ± 5.2	81.4 ± 4.9	0.160
BMI, kg/m ²	23.6 ± 3.4	23.5 ± 3.4	23.5 ± 3.6	23.6 ± 3.7	23.6 ± 3.8	23.0 ± 3.6	0.113
SBP, mmHg	129.8 ± 16.4	127.8 ± 17.4	126.2 ± 16.6	125.7 ± 17.0	125.2 ± 16.6	127.9 ± 17.6	<0.001
DBP, mmHg	68.5 ± 12.0	68.6 ± 11.6	70.8 ± 11.5	72.8 ± 12.1	73.4 ± 13.1	74.7 ± 12.8	<0.001
Creatinine clearance, mL/min	49.5 ± 17.0	49.0 ± 19.3	48.1 ± 17.5	48.1 ± 17.8	47.9 ± 18.0	47.6 ± 18.1	0.691
CHADS ₂ score	3.0 ± 1.2	3.0 ± 1.2	3.0 ± 1.2	2.9 ± 1.2	2.9 ± 1.2	2.7 ± 1.2	<0.001
HAS-BLED score	2.0 ± 0.9	2.0 ± 0.9	1.9 ± 0.9	1.8 ± 0.9	1.8 ± 0.8	1.7 ± 0.8	<0.001
Paroxysmal AF	43 (17.5)	167 (18.6)	669 (16.4)	362 (15.7)	102 (25.6)	153 (42.1)	<0.001
Non-paroxysmal AF	203 (82.5)	733 (81.4)	3410 (83.6)	1943 (84.3)	297 (74.4)	210 (57.9)	
Antiarrhythmic drugs	87 (37.8)	459 (53.6)	2242 (58.0)	1337 (60.4)	236 (62.6)	238 (69.6)	<0.001
Rhythm control	20 (8.7)	85 (9.9)	366 (9.5)	242 (10.9)	57 (15.1)	79 (23.1)	<0.001
Rate control	71 (30.9)	399 (46.6)	2005 (51.9)	1168 (52.8)	192 (50.9)	180 (52.6)	<0.001
Antihypertensive drugs	189 (82.2)	668 (78.0)	3014 (78.0)	1589 (71.8)	277 (73.5)	230 (67.3)	<0.001
Antiplatelet agents	46 (20.0)	159 (18.6)	769 (19.9)	361 (16.3)	61 (16.2)	44 (12.9)	<0.001
DAPT	4 (1.7)	3 (0.4)	27 (0.7)	14 (0.6)	2 (0.5)	0 (0.0)	0.154
ASA	23 (10.0)	91 (10.6)	474 (12.3)	212 (9.6)	37 (9.8)	25 (7.3)	0.006
P2Y12 inhibitors	15 (6.5)	32 (3.7)	172 (4.4)	88 (4.0)	17 (4.5)	10 (2.9)	0.326
Other	14 (6.1)	45 (5.3)	177 (4.6)	96 (4.3)	11 (2.9)	10 (2.9)	0.230
Anticoagulants	241 (98.0)	864 (96.0)	3844 (94.2)	2217 (96.2)	382 (95.7)	338 (93.1)	<0.001
DOACs	170 (69.1)	591 (65.7)	2697 (66.1)	1637 (71.0)	300 (75.2)	271 (74.7)	<0.001
Warfarin	71 (28.9)	272 (30.2)	1147 (28.1)	580 (25.2)	82 (20.6)	67 (18.5)	<0.001
TTR, %	77.2 ± 29.5	77.7 ± 28.4	75.8 ± 30.0	77.5 ± 28.7	73.3 ± 33.4	70.4 ± 30.3	0.463
History of non-pharmacological therapy for AF	12 (4.9)	58 (6.4)	455 (11.2)	194 (8.4)	37 (9.3)	39 (10.7)	<0.001
Catheter ablation	3 (1.2)	25 (2.8)	100 (2.5)	75 (3.3)	22 (5.5)	23 (6.3)	<0.001
Electrical defibrillation	3 (1.2)	10 (1.1)	56 (1.4)	45 (2.0)	10 (2.5)	7 (1.9)	0.210
Implantable cardioverter defibrillator	0 (0.0)	1 (0.1)	18 (0.4)	3 (0.1)	0 (0.0)	0 (0.0)	0.080
Pacemaker	6 (2.4)	25 (2.8)	286 (7.0)	80 (3.5)	11 (2.8)	13 (3.6)	<0.001
Other	0 (0.0)	3 (0.3)	14 (0.3)	10 (0.4)	0 (0.0)	1 (0.3)	0.721
Comorbidities							
Hypertension	194 (78.9)	689 (76.6)	3100 (76.0)	1708 (74.1)	296 (74.2)	267 (73.6)	0.295
Diabetes mellitus	73 (29.7)	253 (28.1)	1164 (28.5)	633 (27.5)	124 (31.1)	93 (25.6)	0.558
Chronic kidney disease	64 (26.0)	211 (23.4)	926 (22.7)	518 (22.5)	88 (22.1)	62 (17.1)	0.133
Myocardial infarction	21 (8.5)	56 (6.2)	221 (5.4)	107 (4.6)	26 (6.5)	10 (2.8)	0.012
Heart failure	106 (43.1)	381 (42.3)	1746 (42.8)	1007 (43.7)	173 (43.4)	123 (33.9)	0.028
History of cerebrovascular disease	55 (22.4)	223 (24.8)	1040 (25.5)	514 (22.3)	84 (21.1)	64 (17.6)	0.002
Gastrointestinal diseases	81 (32.9)	241 (26.8)	1207 (29.6)	692 (30.0)	98 (24.6)	91 (25.1)	0.030
Active cancer	33 (13.4)	83 (9.2)	421 (10.3)	237 (10.3)	32 (8.0)	38 (10.5)	0.316
Dementia	25 (10.2)	93 (10.3)	420 (10.3)	202 (8.8)	33 (8.3)	41 (11.3)	0.285
Fall within 1 year	15 (6.1)	72 (8.0)	305 (7.5)	176 (7.6)	25 (6.3)	28 (7.7)	0.862

Data are n (%) or mean ± standard deviation.

AF, atrial fibrillation; ASA, acetylsalicylic acid; bpm, beats per minute; BMI, body mass index; CHADS₂, congestive heart failure, hypertension, age ≥75 years, diabetes mellitus, stroke; DBP, diastolic blood pressure; DAPT, dual anti-platelet therapy; DOAC, direct oral anticoagulant; HAS-BLED, hypertension, abnormal liver/renal function, stroke history, bleeding history or predisposition, labile prothrombin time international normalized ratio, elderly, drug/alcohol usage; SBP, systolic blood pressure; TTR, time in therapeutic range.

and net clinical outcomes compared with patients taking warfarin. DOACs further reduced the risk of these events compared with warfarin, although statistical significance was not achieved. Baseline resting heart rate did not affect the differences in clinical outcomes between patients treated with DOACs and those treated with warfarin.

There is a high risk of stroke/SEE in patients with AF, which can be predicted by the CHADS₂ score [14]. Furthermore, there have been reports of a higher risk of major bleeding in patients with AF, especially in elderly patients [6,15,16]. The previous ANAFIE sub-analysis assessed the distribution of stroke and bleeding risk scores (55.6 % patients had CHADS₂ score ≥3, indicating a high risk of stroke; 21.1 % had a HAS-BLED score ≥3, indicating an increased risk of bleeding in the Japanese elderly population) [17]. In the present analysis, CHADS₂ and HAS-BLED scores were low in patients with a heart rate ≥110 bpm.

Cardiac complications are among the most common AF complications, affecting 20–30 % of patients with AF [5]. Atrial remodeling, which contributes to the maintenance, progression, and stabilization of AF, is induced by AF itself. Moreover, long-term atrial tachycardia/AF can result in atrial fibrosis contributing to long-term persistence [18]. The cardiac event risk in the present study was high among elderly patients when the heart rate was high. Elderly patients with non-paroxysmal AF and a resting heart rate ≥110 bpm had a 1.8-fold higher

cardiac event risk than those with heart rate 60 to <80 bpm; there was no significant difference in the occurrence of CV events in patients with low heart rate (<60 bpm). We also noted that CV death and all-cause death tended to be higher with a heart rate ≥110 bpm. Because cardiac events are important as an event leading to CV death or all-cause death, more careful management of heart disease in elderly NVAf patients with a high heart rate is warranted.

Our results are consistent with those of a study from the Fushimi AF registry, which demonstrated a higher incidence of adverse events (a composite endpoint of all-cause death, hospitalization for HF, stroke/SEE, myocardial infarction, and tachycardia- or bradycardia-related events) in Japanese patients with sustained AF and with a resting heart rate ≥110 bpm compared with 60–79 bpm. In patients with paroxysmal AF, no association between heart rate and adverse events was observed [19]. In addition, the findings of the present study are in line with results from the RACE-II trial, which showed no difference in the occurrence of a composite endpoint (death from CV causes, hospitalization for HF, SEE, bleeding, and life-threatening arrhythmic events) between lenient- (<110 bpm) and strict- (<80 bpm) rate control groups [7]. In the ORBIT-AF study, a J-shaped relationship between heart rate and all-cause death was noted, wherein both a decreasing heart rate of ≤65 bpm (aHR, 1.15 per 5-bpm decrease; 95 % CI,

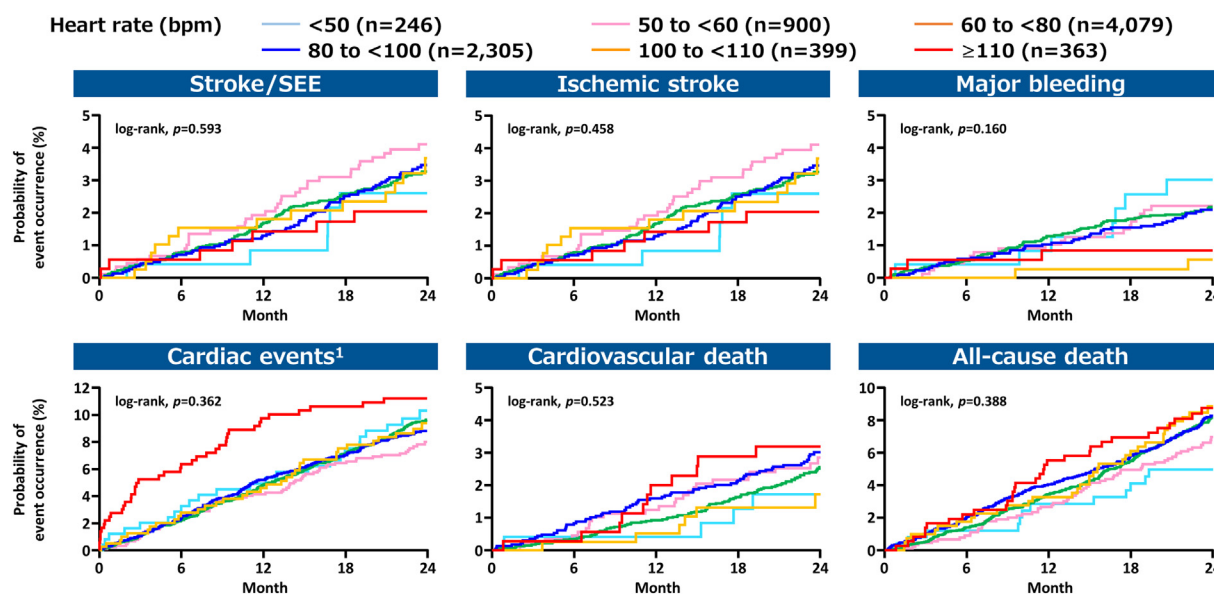


Fig. 2. Kaplan-Meier curves for clinical outcomes by heart rate.

¹Cardiac events were a composite of myocardial infarction, cardiac intervention for ischemic disease other than myocardial infarction, heart failure requiring hospitalization, and cardiovascular death.

bpm, beats per minute; SEE, systemic embolic event.

1.01–1.32; $p = 0.04$) and increasing heart rate >65 bpm (aHR, 1.10 per 5-bpm increase; 95% CI, 1.05–1.15; $p < 0.0001$) were associated with an increased risk of all-cause death [6]. This is inconsistent with results from another study, which indicated no relationship between heart rate and all-cause death in AF patients but an association between heart rate and hospitalization due to heart disease [9]. In the present study, there was a tendency for all-cause death to increase with a heart rate ≥ 110 bpm compared with heart rate 60 to <80 bpm, although this was only statistically significant in patients with non-paroxysmal AF. This is also the first study to evaluate the relationship between heart rate and risk of stroke/SEE.

The present analysis indicated that elderly patients with a heart rate of 60 to <80 bpm who were not using OACs had a 2-fold significantly higher risk of developing stroke/SEE and a 2.5-fold higher risk of ischemic stroke than those using warfarin. Concordant findings were reported in the main analysis of the ANAFIE Registry, which indicated that when compared with patients on warfarin, patients taking DOACs

had a lower risk for stroke/SEE, major bleeding, and all-cause death after adjusting for confounders, whereas patients in the No-OAC group had a higher risk for stroke/SEE and all-cause death and a lower risk for major bleeding [20].

There is a need for appropriate use of OACs among elderly patients. The present analysis showed consistent HRs for clinical outcomes for patients taking DOACs vs warfarin in all heart rate subgroups. This highlights the importance of appropriate OAC use as recommended in guidelines, irrespective of heart rate during AF. As stroke/SEE risk was observed to be high in patients with a heart rate of 60 to <80 bpm who were not taking any OACs, appropriate OAC therapy is considered necessary for these patients. The primary goal of heart rate control in AF is to minimize symptoms associated with tachycardia and prevent subsequent cardiomyopathy. Additionally, rate reduction may also lead to improved hemodynamic status by allowing adequate time for ventricular filling and avoiding rate-related ischemia. Furthermore, strict heart rate control attenuates prothrombotic state and platelet activity in

Table 2

Incidence rates for each clinical outcome stratified by baseline heart rate in AF.

Incidence rate (95% CI), 100 persons year	Heart rate (bpm), n = 8292					
	<50 (n = 246)	50 to <60 (n = 900)	60 to <80 (n = 4079)	80 to <100 (n = 2305)	100 to <110 (n = 399)	≥ 110 (n = 363)
Stroke/SEE	1.31 (0.26–2.37)	2.10 (1.40–2.79)	1.66 (1.37–1.95)	1.74 (1.35–2.14)	1.78 (0.81–2.75)	1.05 (0.27–1.84)
Ischemic stroke	0.66 (0.00–1.40)	1.67 (1.05–2.29)	1.25 (0.99–1.50)	1.44 (1.08–1.80)	1.64 (0.71–2.57)	1.05 (0.27–1.84)
Major bleeding	1.53 (0.40–2.66)	1.13 (0.62–1.64)	1.10 (0.86–1.34)	1.06 (0.75–1.37)	0.27 (0.00–0.65)	0.45 (0.00–0.96)
Cardiovascular events ^a	6.42 (4.04–8.79)	5.72 (4.56–6.88)	6.44 (5.85–7.02)	5.90 (5.16–6.65)	6.41 (4.53–8.28)	7.21 (5.10–9.32)
Cardiac events ^b	5.45 (3.27–7.63)	4.19 (3.20–5.17)	5.08 (4.57–5.60)	4.70 (4.04–5.36)	4.93 (3.30–6.57)	6.35 (4.38–8.32)
Myocardial infarction	0.00 (0.00–0.00)	0.12 (0.00–0.28)	0.28 (0.16–0.39)	0.33 (0.16–0.50)	0.00 (0.00–0.00)	0.45 (0.00–0.96)
Cardiac intervention for ischemic disease other than myocardial infarction	0.22 (0.00–0.64)	0.36 (0.07–0.64)	0.29 (0.17–0.41)	0.16 (0.04–0.28)	0.27 (0.00–0.65)	0.30 (0.00–0.71)
Heart failure requiring hospitalization	5.22 (3.09–7.35)	3.81 (2.87–4.75)	4.66 (4.17–5.15)	4.39 (3.76–5.03)	4.93 (3.29–6.56)	6.18 (4.24–8.11)
Cardiovascular death	0.87 (0.02–1.72)	1.42 (0.85–1.98)	1.26 (1.01–1.51)	1.52 (1.15–1.89)	0.81 (0.16–1.46)	1.64 (0.67–2.62)
All-cause death	2.61 (1.13–4.08)	3.54 (2.64–4.44)	4.22 (3.76–4.68)	4.30 (3.67–4.92)	4.60 (3.06–6.15)	4.63 (3.00–6.26)
Net clinical outcome ^c	4.19 (2.30–6.07)	5.97 (4.80–7.15)	6.12 (5.56–6.69)	6.11 (5.36–6.86)	6.32 (4.49–8.14)	5.43 (3.65–7.20)

AF, atrial fibrillation; bpm, beats per minute; SEE, systemic embolic event.

^a Cardiovascular events were a composite of stroke and cardiac events.

^b Cardiac events were a composite of myocardial infarction, cardiac intervention for ischemic disease other than myocardial infarction, heart failure requiring hospitalization, and cardiovascular death.

^c Net clinical outcome was a composite of stroke/SEE, major bleeding, and all-cause death.

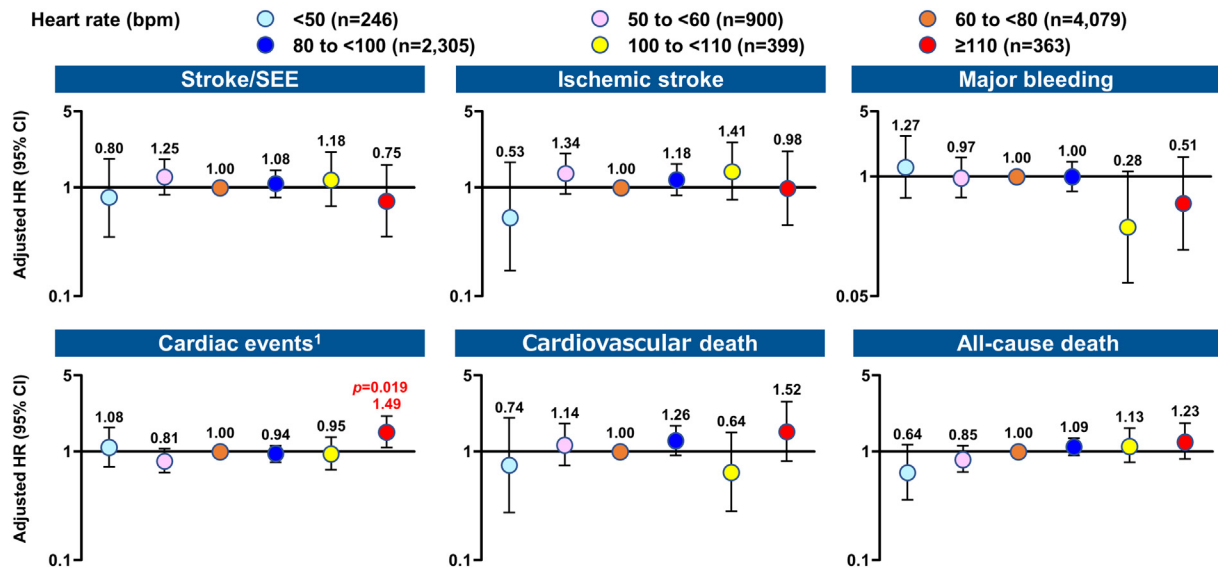


Fig. 3. Adjusted hazard ratio for each clinical outcome by heart rate.

Reference: 60 to <80 bpm.

¹Cardiac events were a composite of myocardial infarction, cardiac intervention for ischemic disease other than myocardial infarction, heart failure requiring hospitalization, and cardiovascular death.

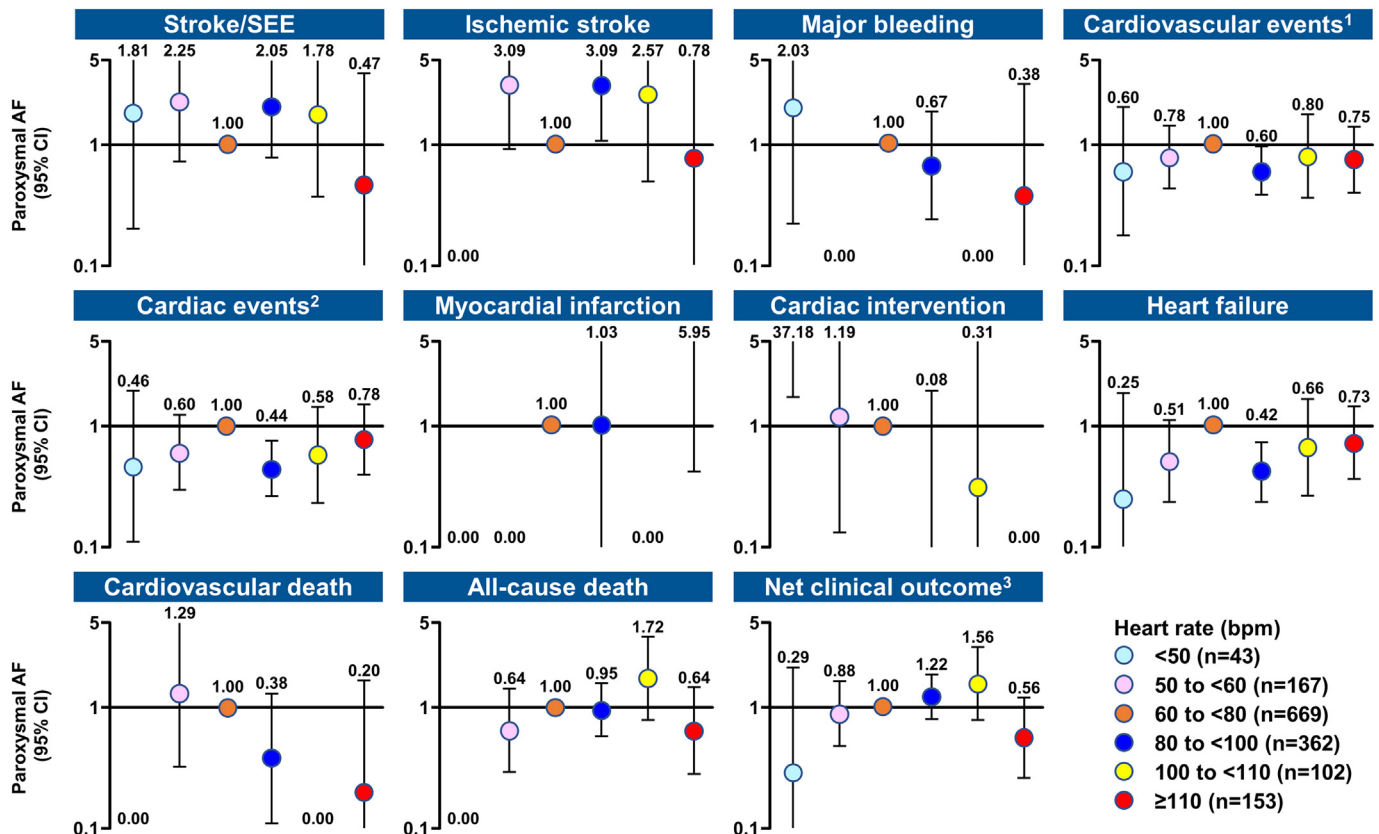
bpm, beats per minute; CI, confidence interval; HR, hazard ratio; SEE, systemic embolic event.

NVAF [21], demonstrating the importance of reducing resting heart rate during AF in patients with heart rate ≥110 bpm.

Limitations

The limitations of the ANAFIE Registry have been previously described [17,20] and are primarily related to the observational study design and

inclusion of only Japanese patients, thereby limiting the generalizability of the findings. In addition, although the data were adjusted for known confounders, unknown confounding factors cannot be completely ruled out. Additionally, we could not assess the impact of OACs among patients in other heart rate categories, such as <60 bpm or ≥80 bpm, because there were few patients in these heart rate categories, which limited the analysis and conclusions to be drawn. Data on prior coronary interventions



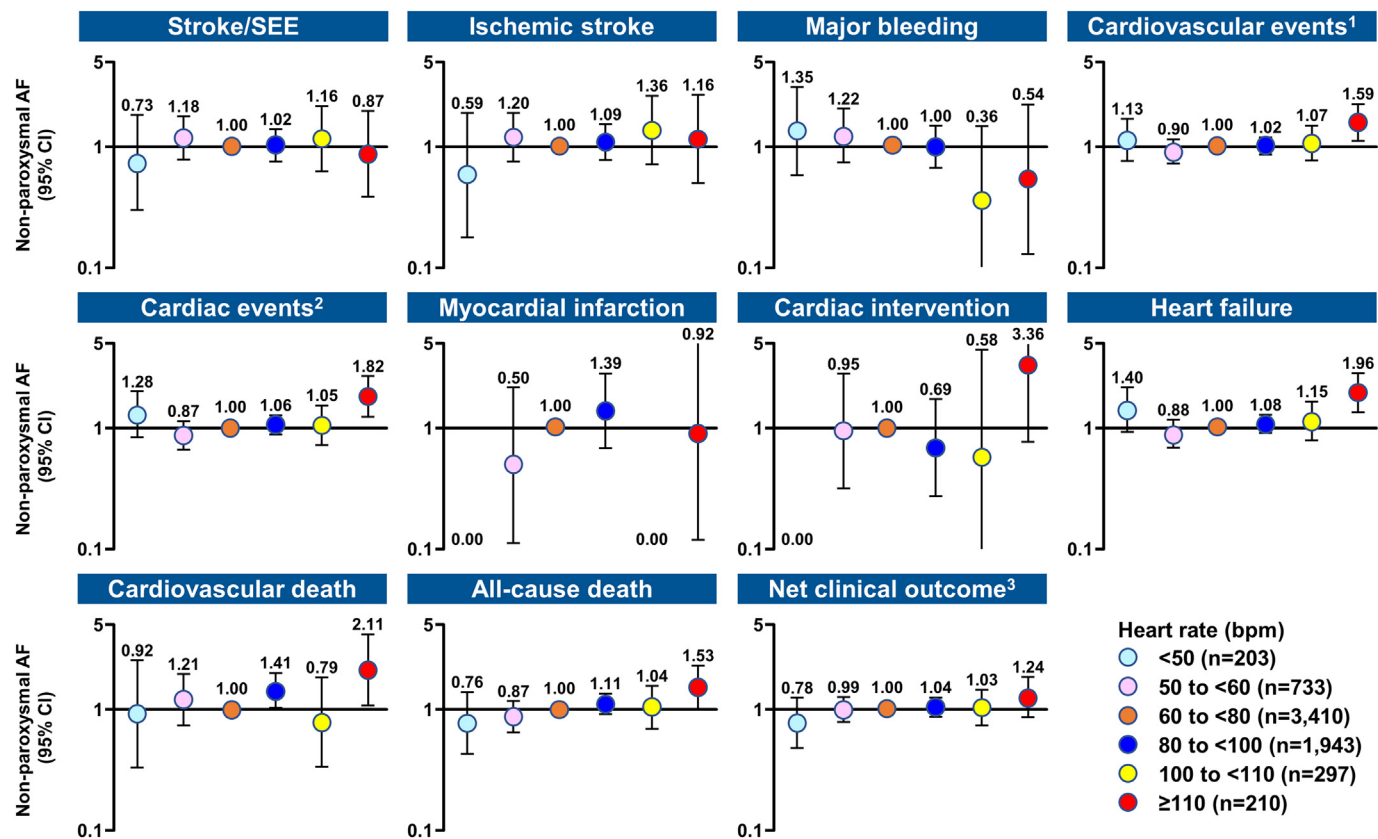


Fig. 4. Adjusted hazard ratio for each clinical outcome by heart rate in paroxysmal AF and non-paroxysmal AF groups.

Reference: 60 to <80 bpm.

¹Cardiovascular events were a composite of stroke and cardiac events. ²Cardiac events were a composite of myocardial infarction, cardiac intervention for ischemic disease other than myocardial infarction, heart failure requiring hospitalization, and cardiovascular death. ³Net clinical outcome was a composite of stroke/SEE, major bleeding, and all-cause death.

*The p-value of the interaction between heart rate and the type of AF.

Sex, age, body mass index, history of major bleeding, hypertension, severe hepatic dysfunction, diabetes mellitus, hyperuricemia, heart disease (heart failure, left ventricular systolic dysfunction), myocardial infarction, cerebrovascular disease, thromboembolic-related disease, active cancer, dementia, falls within 1 year, anticoagulants, catheter ablation, antiarrhythmics, antiplatelet agents, proton pump inhibitors, P-glycoprotein inhibitors, dyslipidemia, creatinine clearance, gastrointestinal disease, and polypharmacy were included in the model. AF, atrial fibrillation; bpm, beats per minute; CI, confidence interval; HR, hazard ratio; SEE, systemic embolic event.

were not obtained in the ANAFIE Registry; however, prior coronary interventions may potentially affect the occurrence of cardiac events. Finally, as this subgroup analysis was based on heart rate data obtained at the time of enrollment, it did not account for heart rate fluctuations.

Conclusions

The results of this sub-cohort study indicated an association between a heart rate ≥110 bpm and an increased risk of cardiac events

Table 3

Adjusted HRs of DOAC use and No-OAC use compared with warfarin among patients with resting heart rate of 60 to <80 bpm in AF.

Heart rate 60 to <80 bpm (n = 4079)	DOAC (n = 2697)		No-OACs (n = 235)	
	HR (95 % CI)	p-Value	HR (95 % CI)	p-Value
Stroke/SEE	0.72 (0.48–1.07)	0.100	2.08 (1.06–4.07)	0.033
Ischemic stroke	0.70 (0.45–1.11)	0.130	2.57 (1.23–5.36)	0.012
Major bleeding	0.75 (0.47–1.22)	0.247	1.40 (0.58–3.42)	0.455
Cardiovascular events ^a	0.95 (0.77–1.17)	0.630	1.43 (0.98–2.08)	0.060
Cardiac events ^b	0.99 (0.79–1.25)	0.938	1.23 (0.80–1.89)	0.352
Myocardial infarction	0.67 (0.26–1.69)	0.393	0.40 (0.04–3.91)	0.433
Cardiac intervention for ischemic disease other than myocardial infarction	1.40 (0.48–4.04)	0.537	1.93 (0.31–12.10)	0.481
Heart failure requiring hospitalization	0.97 (0.77–1.23)	0.814	1.25 (0.80–1.95)	0.326
Cardiovascular death	0.91 (0.57–1.44)	0.685	1.75 (0.83–3.68)	0.143
All-cause death	0.95 (0.74–1.22)	0.676	1.45 (0.94–2.23)	0.091
Net clinical outcome ^c	0.86 (0.70–1.06)	0.162	1.62 (1.13–2.33)	0.009

Confounding variables included in the model are as listed in Table 2 and type of AF.

AF, atrial fibrillation; bpm, beats per minute; CI, confidence interval; DOAC, direct oral anticoagulant; HR, hazard ratio; No-OAC, no-use of oral anticoagulants; SEE, systemic embolic event.

^a Cardiovascular events.

^b Cardiac events.

^c Net clinical outcomes are as in Table 2.

in elderly patients with non-paroxysmal NVAf. Low heart rate in elderly patients with AF did not affect the occurrence of events. The study also showed that the incidence of adverse clinical outcomes was similar across heart rate groups in patients taking DOACs compared with those taking warfarin. In elderly patients with AF and with a heart rate of 60 to <80 bpm, OACs decreased the incidence of stroke/SEE and ischemic stroke.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcc.2022.11.011>.

CRedit authorship contribution statement

TI, T Yamashita, MA, HA, YK, KO, WS, SS, HT, KT, AH, MY, T Yamaguchi, and HI designed and conducted the study; TI interpreted the data analysis; ST carried out the statistical analyses; TI, TK, YM, and AT wrote and reviewed the manuscript; all authors revised and commented on the manuscript, and approved the final version.

Data availability

The datasets used in the current analysis are available from the corresponding author upon reasonable request and after review by a committee led by the study sponsor.

Declaration of competing interest

Takanori Ikeda received research grants from Daiichi Sankyo, Medtronic Japan, and Japan Lifeline and honoraria from Ono Pharma, Bayer Yakuhin, Daiichi Sankyo, Bristol-Myers Squibb, and Pfizer. Additionally, Takanori Ikeda was a member of the advisory board for Bayer Yakuhin, Bristol-Myers Squibb, and Daiichi Sankyo.

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Tetsuya Kimura has stock and is an employee of Daiichi Sankyo.

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